事件與事故回應

Incident & Event Response

AWS Certified DevOps Engineer – Professional (DOP-C02) ｜ 官方考試權重 14%

第 1 題 （題庫 No.9 · 複選（Select 2））

A company uses an organization in AWS Organizations to manage multiple AWS accounts. The company needs a solution to detect sensitive information in Amazon S3 buckets in all the company's accounts. When the solution detects sensitive data, the solution must collect all the findings and make them available to the company's security officer in a single location. The solution must move S3 objects that contain sensitive information to a quarantine S3 bucket. Which solutions will meet these requirements with the LEAST operational overhead? (Select TWO.)

A. Enable AWS Security Hub in the organization. Enable Amazon Macie for all the accounts in the organization. Configure Macie to send findings to Security Hub.

B. Create an AWS Service Catalog product to provision S3 buckets. Configure Service Catalog to create a new S3 bucket. Configure S3 Event Notifications to send ObjectCreated events to an Amazon Simple Queue Service (Amazon SQS) queue.

C. Create an AWS Lambda function to copy S3 objects from S3 buckets to a dedicated quarantine bucket. Configure the Lambda function to delete copied objects from the original buckets. Configure an Amazon EventBridge rule to invoke the Lambda function in response to sensitive information findings from Amazon Macie.

D. Configure an AWS Lambda function to run when new objects are created or when existing objects are updated. Configure the Lambda function to determine whether objects contain sensitive data. Configure the function to move objects that contain sensitive data to a quarantine bucket and to delete the original objects.

E. Configure SCPs to prevent the creation of S3 buckets and objects that contain suspected sensitive data.Configure the SCPs to move objects that are suspected to contain sensitive data to a dedicated quarantine S3 bucket.

我的答案：_____

第 2 題 （題庫 No.11 · 單選）

A company has a public application that uses an Amazon API Gateway REST API, an AWS Lambda function, and an Amazon RDS for PostgreSQL DB cluster. Users have recently received error messages as application demand increased. The company ' s DevOps engineer discovered that the errors were caused by RDS connection limits being reached. The DevOps engineer also discovered that more than 90% of the API requests are GET requests that read from the DB cluster. How should the DevOps engineer solve this problem with the LEAST development effort?

A. Migrate from Amazon RDS to Amazon DynamoDB. Add an Amazon CloudFront distribution in front of the API Gateway REST API.

B. Add a proxy from Amazon RDS Proxy in front of the RDS DB cluster. Enable API caching in API Gateway.

C. Add an Amazon RDS Proxy in front of the RDS database cluster. Provision an Amazon ElastiCache (Redis OSS) cluster.

D. Migrate from Amazon RDS to Amazon DynamoDB. Enable API caching in API Gateway.

我的答案：_____

第 3 題 （題庫 No.63 · 單選）

A company runs hundreds of EC2 instances with new instances launched/terminated hourly. Security requires all running instances to have an instance profile attached. A default profile exists and must be attached automatically to any instance missing one. Which solution meets this requirement?

A. EventBridge rule for RunInstances API calls, invoke Lambda to attach default profile.

B. AWS Config with ec2-instance-profile-attached managed rule, automatic remediation using Systems Manager Automation runbook to attach profile.

C. EventBridge rule for StartInstances API calls, invoke Systems Manager Automation runbook to attach profile.

D. AWS Config iam-role-managed-policy-check managed rule, automatic remediation with Lambda to attach profile.

我的答案：_____

第 4 題 （題庫 No.73 · 複選（Select 2））

A company ' s DevOps engineer is creating an AWS Lambda function to process notifications from an Amazon Simple Notification Service (Amazon SNS) topic. The Lambda function will process the notification messages and will write the contents of the notification messages to an Amazon RDS Multi-AZ DB instance. During testing a database administrator accidentally shut down the DB instance. While the database was down the company lost several of the SNS notification messages that were delivered during that time. The DevOps engineer needs to prevent the loss of notification messages in the future Which solutions will meet this requirement? (Select TWO.)

A. Replace the RDS Multi-AZ DB instance with an Amazon DynamoDB table.

B. Configure an Amazon Simple Queue Service (Amazon SQS) queue as a destination of the Lambda function.

C. Configure an Amazon Simple Queue Service (Amazon SQS > dead-letter queue for the SNS topic.

D. Subscribe an Amazon Simple Queue Service (Amazon SQS) queue to the SNS topic Configure the Lambda function to process messages from the SQS queue.

E. Replace the SNS topic with an Amazon EventBridge event bus Configure an EventBridge rule on the new event bus to invoke the Lambda function for each event.

我的答案：_____

第 5 題 （題庫 No.83 · 單選）

A company uses Amazon RDS for all databases in Its AWS accounts The company uses AWS Control Tower to build a landing zone that has an audit and logging account All databases must be encrypted at rest for compliance reasons. The company ' s security engineer needs to receive notification about any noncompliant databases that are in the company ' s accounts Which solution will meet these requirements with the MOST operational efficiency?

A. Use AWS Control Tower to activate the optional detective control (guardrail) to determine whether the RDS storage is encrypted Create an Amazon Simple Notification Service (Amazon SNS) topic in the company ' s audit account. Create an Amazon EventBridge rule to filter noncompliant events from the AWS Control Tower control (guardrail) to notify the SNS topic. Subscribe the security engineer ' s email address to the SNS topic

B. Use AWS Cloud Formation StackSets to deploy AWS Lambda functions to every account. Write the Lambda function code to determine whether the RDS storage is encrypted in the account the function is deployed to Send the findings as an Amazon CloudWatch metric to the management account Create an Amazon Simple Notification Service (Amazon SNS) topic. Create a CloudWatch alarm that notifies the SNS topic when metric thresholds are met. Subscribe the security engineer ' s email address to the SNS topic.

C. Create a custom AWS Config rule in every account to determine whether the RDS storage is encrypted Create an Amazon Simple Notification Service (Amazon SNS) topic in the audit account Create an Amazon EventBridge rule to filter noncompliant events from the AWS Control Tower control (guardrail) to notify the SNS topic. Subscribe the security engineer ' s email address to the SNS topic

D. Launch an Amazon EC2 instance. Run an hourly cron job by using the AWS CLI to determine whether the RDS storage is encrypted in each AWS account Store the results in an RDS database. Notify the security engineer by sending email messages from the EC2 instance when noncompliance is detected

我的答案：_____

第 6 題 （題庫 No.168 · 單選）

A company has a file-reading application that saves files to a database running on Amazon EC2 instances. Regulations require daily file deletions from EC2 instances and deletion of database records older than 60 days. Database record deletion must occur after file deletion. The company needs email notifications for any deletion script failures. Which solution will meet these requirements with the LEAST development effort?

A. Use AWS Systems Manager State Manager to automatically invoke an Automation document at the specified time daily. Configure the Automation document to run deletion scripts sequentially via run command. Create an EventBridge rule to send failure notifications to Amazon SNS.

B. Use AWS Systems Manager State Manager to automatically invoke an Automation document at the specified time daily. Configure the Automation document to run deletion scripts sequentially. Add a conditional check for errors as the last step and send failure notifications via Amazon SES.

C. Create an EventBridge rule to invoke a Lambda function at the specified time. Configure the Lambda function to run deletion scripts sequentially and send failure notifications via SNS.

D. Create an EventBridge rule to invoke a Lambda function at the specified time. Configure the Lambda function to run deletion scripts sequentially and send failure notifications via SES.

我的答案：_____

第 7 題 （題庫 No.217 · 複選（Select 2））

A company has an application that stores data that includes personally Identifiable Information (Pll) In an Amazon S3 bucket All data Is encrypted with AWS Key Management Service (AWS KMS) customer managed keys. All AWS resources are deployed from an AWS Cloud Formation template. A DevOps engineer needs to set up a development environment for the application in a different AWS account The data in the development environment ' s S3 bucket needs to be updated once a week from the production environment ' s S3 bucket. The company must not move Pll from the production environment without anonymizmg the Pll first The data in each environment must be encrypted with different KMS customer managed keys. Which combination of steps should the DevOps engineer take to meet these requirements? (Select TWO)

A. Activate Amazon Macie on the S3 bucket In the production account Create an AWS Step Functions state machine to initiate a discovery job and redact all Pll before copying files to the S3 bucket in the development account. Give the state machine tasks decrypt permissions on the KMS key in the production account. Give the state machine tasks encrypt permissions on the KMS key in the development account

B. Set up S3 replication between the production S3 bucket and the development S3 bucket Activate Amazon Macie on the development S3 bucket Create an AWS Step Functions state machine to initiate a discovery job and redact all Pll as the files are copied to the development S3 bucket. Give the state machine tasks encrypt and decrypt permissions on the KMS key in the development account.

C. Set up an S3 Batch Operations job to copy files from the production S3 bucket to the development S3 bucket. In the development account, configure anAWS Lambda function to redact all Pll. Configure S3 Object Lambda to use the Lambda function for S3 GET requests Give the Lambda function ' s 1AM role encrypt and decrypt permissions on the KMS key in the development account.

D. Create a development environment from the CloudFormatlon template in the development account. Schedule an Amazon EventBridge rule to start the AWS Step Functions state machine once a week

E. Create a development environment from the CloudFormation template in the development account.Schedule a cron job on an Amazon EC2 instance to run once a week to start the S3 Batch Operations job.

我的答案：_____

第 8 題 （題庫 No.274 · 單選）

A company has an application that runs on a fleet of Amazon EC2 instances. The application requires frequent restarts. The application logs contain error messages when a restart is required. The application logs are published to a log group in Amazon CloudWatch Logs. An Amazon CloudWatch alarm notifies an application engineer through an Amazon Simple Notification Service (Amazon SNS)

topic when the logs contain a large number of restart-related error messages. The application engineer manually restarts the application on the instances after the application engineer receives a notification from the SNS topic. A DevOps engineer needs to implement a solution to automate the application restart on the instances without restarting the instances. Which solution will meet these requirements in the MOST operationally efficient manner?

A. Configure an AWS Systems Manager Automation runbook that runs a script to restart the application on the instances. Configure the SNS topic to invoke the runbook.

B. Create an AWS Lambda function that restarts the application on the instances. Configure the Lambda function as an event destination of the SNS topic.

C. Configure an AWS Systems Manager Automation runbook that runs a script to restart the application on the instances. Create an AWS Lambda function to invoke the runbook. Configure the Lambda function as an event destination of the SNS topic.

D. Configure an AWS Systems Manager Automation runbook that runs a script to restart the application on the instances. Configure an Amazon EventBridge rule that reacts when the CloudWatch alarm enters ALARM state. Specify the runbook as a target of the rule.

我的答案：_____

第 9 題 （題庫 No.311 · 單選）

A media company has several thousand Amazon EC2 instances in an AWS account. The company is using Slack and a shared email inbox for team communications and important updates. A DevOps engineer needs to send all AWS-scheduled EC2 maintenance notifications to the Slack channel and the shared inbox. The solution must include the instances ' Name and Owner tags. Which solution will meet these requirements?

A. Integrate AWS Trusted Advisor with AWS Config Configure a custom AWS Config rule to invoke an AWS Lambda function to publish notifications to an Amazon Simple Notification Service (Amazon SNS) topic Subscribe a Slack channel endpoint and the shared inbox to the topic.

B. Use Amazon EventBridge to monitor for AWS Health Events Configure the maintenance events to target an Amazon Simple Notification Service (Amazon SNS) topic Subscribe an AWS Lambda function to the SNS topic to send notifications to the Slack channel and the shared inbox.

C. Create an AWS Lambda function that sends EC2 maintenance notifications to the Slack channel and the shared inbox Monitor EC2 health events by using Amazon CloudWatch metrics Configure a CloudWatch alarm that invokes the Lambda function when a maintenance notification is received.

D. Configure AWS Support integration with AWS CloudTrail Create a CloudTrail lookup event to invoke an AWS Lambda function to pass EC2 maintenance notifications to Amazon Simple Notification Service (Amazon SNS) Configure Amazon SNS to target the Slack channel and the shared inbox.

我的答案：_____

第 10 題 （題庫 No.379 · 單選）

A company sends its AWS Network Firewall flow logs to an Amazon S3 bucket. The company then analyzes the flow logs by using Amazon Athena. The company needs to transform the flow logs and add additional data before the flow logs are delivered to the existing S3 bucket. Which solution will meet these requirements?

A. Create an AWS Lambda function to transform the data and to write a new object to the existing S3 bucket. Configure the Lambda function with an S3 trigger for the existing S3 bucket. Specify all object create events for the event type. Acknowledge the recursive invocation.

B. Enable Amazon EventBridge notifications on the existing S3 bucket. Create a custom EventBridge event bus. Create an EventBridge rule that is associated with the custom event bus. Configure the rule to react to all object create events for the existing S3 bucket and to invoke an AWS Step Functions workflow. Configure a Step Functions task to transform the data and to write the data into a new S3 bucket.

C. Create an Amazon EventBridge rule that is associated with the default EventBridge event bus. Configure the rule to react to all object create events for the existing S3 bucket. Define a new S3

bucket as the target for the rule. Create an EventBridge input transformation to customize the event before passing the event to the rule target.

D. Create an Amazon Data Firehose delivery stream that is configured with an AWS Lambda transformer.Specify the existing S3 bucket as the destination. Change the Network Firewall logging destination from Amazon S3 to Firehose.

我的答案：_____

解答與解析　Answer Key

解析取自題庫原文，未經 AWS 官方審核；若與官方文件有出入，以官方文件為準。

第 1 題（No.9）　正確答案：A、C

The company requires an organization-wide, centralized, and automated solution to detect sensitive data in Amazon S3, aggregate findings in one location, and quarantine affected objects with minimal operational overhead. AWS provides a native service specifically designed for this purpose: Amazon Macie .

Option A is essential because Amazon Macie automatically discovers and classifies sensitive data such as PII in S3 buckets using managed machine learning models. When enabled at the organization level , Macie scans buckets across all accounts and Regions. Integrating Macie with AWS Security Hub centralizes all findings in a single dashboard, allowing the company's security officer to review and manage sensitive data alerts across the organization without building custom aggregation pipelines.

Detection alone is not sufficient; remediation is also required. Option C completes the solution by using Amazon EventBridge , which natively receives Macie findings in near real time. An EventBridge rule can trigger a Lambda function whenever Macie identifies sensitive data. The Lambda function can then copy the affected object to a quarantine S3 bucket and delete the original object, meeting the remediation requirement automatically and consistently.

Option D requires custom sensitive data detection logic, which is complex, error-prone, and unnecessary given Macie's capabilities. Option E is invalid because SCPs cannot inspect or move data. Option B does not detect sensitive information.

Therefore, A and C together provide the most efficient, scalable, and AWS-recommended solution.

涉及服務：Lambda、Organizations/Control Tower、S3、SQS、EventBridge、Security Hub

第 2 題（No.11）　正確答案：B

The root cause of the problem is that the RDS PostgreSQL instance is exceeding its database connection limit due to high volumes of concurrent GET requests. The most efficient and lowest-effort remediation is to reduce the number of connections that reach the database and serve as many reads as possible from a cache layer. Option B accomplishes both goals with minimal changes to the existing architecture.

First, RDS Proxy manages database connections efficiently by pooling and reusing them. This reduces connection churn and prevents the Lambda function from opening excessive concurrent connections during traffic spikes. RDS Proxy is natively integrated with RDS and requires only minor configuration changes without altering application logic.

Second, enabling API Gateway caching allows caching GET responses directly at the API layer. Because over

90% of requests are GET operations, enabling caching drastically reduces traffic hitting both Lambda and RDS, significantly improving performance and lowering connection pressure.

Option C also improves read scaling, but adding ElastiCache introduces additional infrastructure and requires modifying application code to query Redis before querying the database. This is more complex than enabling API Gateway caching. Options A and D require full database migration, which is far more labor-intensive.

Option B is the simplest, most effective, and most aligned with AWS best practices for reducing RDS connection saturation.

涉及服務：Lambda、RDS/Aurora、DynamoDB、API Gateway

第 3 題（No.63）　正確答案：B

* AWS Config's ec2-instance-profile-attached managed rule checks for attached instance profiles.
* Config supports automatic remediation via Systems Manager Automation runbooks.

* This provides continuous compliance with minimal operational overhead.
* EventBridge and Lambda (A) require custom coding and risk missing existing instances.
* StartInstances (C) does not cover RunInstances and new instances.
* IAM-role managed policy check (D) does not check instance profile attachments.
References:
AWS Config Managed Rules
Config Automatic Remediation
涉及服務：Systems Manager (SSM)、Lambda、EC2/Auto Scaling、Config、IAM/SCP、EventBridge

第 4 題（No.73） 正確答案：C、D

These solutions will meet the requirement because they will prevent the loss of notification messages in the future. An Amazon SQS queue is a service that provides a reliable, scalable, and secure message queue for asynchronous communication between distributed components. You can use an SQS queue to buffer messages from an SNS topic and ensure that they are delivered and processed by a Lambda function, even if the function or the database is temporarily unavailable.
Option C will configure an SQS dead-letter queue for the SNS topic. A dead-letter queue is a queue that receives messages that could not be delivered to any subscriber after a specified number of retries. You can use a dead-letter queue to store and analyze failed messages, or to reprocess them later. This way, you can avoid losing messages that could not be delivered to the Lambda function due to network errors, throttling, or other issues.
Option D will subscribe an SQS queue to the SNS topic and configure the Lambda function to process messages from the SQS queue. This will decouple the SNS topic from the Lambda function and provide more flexibility and control over the message delivery and processing. You can use an SQS queue to store messages from the SNS topic until they are ready to be processed by the Lambda function, and also to retry processing in case of failures. This way, you can avoid losing messages that could not be processed by the Lambda function due to database errors, timeouts, or other issues.
涉及服務：Lambda、RDS/Aurora、DynamoDB、SNS、SQS、EventBridge

第 5 題（No.83） 正確答案：A

Activate AWS Control Tower Guardrail:
Use AWS Control Tower to activate a detective guardrail that checks whether RDS storage is encrypted.
Create SNS Topic for Notifications:
Set up an Amazon Simple Notification Service (SNS) topic in the audit account to receive notifications about non-compliant databases.
Create EventBridge Rule to Filter Non-compliant Events:
Create an Amazon EventBridge rule that filters events related to the guardrail ' s findings on non-compliant RDS instances.
Configure the rule to send notifications to the SNS topic when non-compliant events are detected.
Subscribe Security Engineer ' s Email to SNS Topic:
Subscribe the security engineer ' s email address to the SNS topic to receive notifications when non-compliant databases are detected.
By using AWS Control Tower to activate a detective guardrail and setting up SNS notifications for non- compliant events, the company can efficiently monitor and ensure that all RDS databases are encrypted at rest.
References:
AWS Control Tower Guardrails
Amazon SNS
Amazon EventBridge
涉及服務：Lambda、EC2/Auto Scaling、CloudWatch、Config、Organizations/Control Tower、RDS/Aurora

第 6 題（No.168） 正確答案：A

* Systems Manager State Manager with Automation documents allows running scripts sequentially and reliably with built-in retry and status tracking.
* Using EventBridge with SNS for notifications leverages managed services with minimal custom development.
* Using SES (Options B, D) requires more setup and custom logic for email formatting and sending.
* Lambda (Options C, D) can run scripts but might have limitations on execution time and complexity compared to Systems Manager Automation.

References:

AWS Systems Manager Automation

Monitoring Automation Execution and Notifications

涉及服務：Systems Manager (SSM)、Lambda、EC2/Auto Scaling、SNS、EventBridge

第 7 題（No.217） 正確答案：A、D

Activate Amazon Macie on the Production S3 Bucket:

Macie can identify and protect sensitive data such as PII.

Create a Step Functions state machine to automate data discovery and redaction before copying it to the development environment.

Example Step Functions state machine:

{
" Comment " : " Anonymize PII and copy data " ,
" StartAt " : " MacieDiscoveryJob " ,
" States " : {
" MacieDiscoveryJob " : {
" Type " : " Task " ,
" Resource " : " arn:aws:states:::macie:startClassificationJob " ,
" End " : true
}
}
}

Create a Development Environment from CloudFormation Template:

Deploy the development environment in a new account using the existing CloudFormation template.

Schedule an EventBridge rule to start the Step Functions state machine on a weekly basis.

EventBridge rule example:

{
" ScheduleExpression " : " rate(7 days) " ,
" StateMachineArn " : " arn:aws:states: < region > : < account-id >
:stateMachine:AnonymizeAndCopyData "
}

By using Macie for data anonymization and Step Functions for automation, you ensure PII is properly handled before data transfer between environments.

References:

Amazon Macie

AWS Step Functions

AWS CloudFormation Templates

涉及服務：CloudFormation、Lambda、EC2/Auto Scaling、KMS、S3、EventBridge

第 8 題（No.274） 正確答案：D

This solution meets the requirements in the most operationally efficient manner by automating the application restart process on the instances without restarting them. When the CloudWatch alarm

enters the ALARM state, the EventBridge rule is triggered, which in turn invokes the Systems Manager Automation runbook that contains the script to restart the application on the instances.

涉及服務：Systems Manager (SSM)、Lambda、EC2/Auto Scaling、CloudWatch、SNS、EventBridge

第 9 題（No.311） 正確答案：B

https://docs.aws.amazon.com/health/latest/ug/cloudwatch-events-health.html

涉及服務：Lambda、EC2/Auto Scaling、CloudWatch、CloudTrail、Config、SNS

第 10 題（No.379） 正確答案：D

（此題題庫未附解析）

涉及服務：Lambda、S3、EventBridge、Step Functions、Athena